

RP2350 Split Keyboard Firmware

A modular, C++ based firmware for an RP2350-powered split keyboard. This project integrates multiple hardware subsystems into a unified, non-blocking architecture.

Diagrams

Here is a link to the [diagrams](#).

For more information, see the the [docs](#) directory.

Features

- **ADC Joystick Interface:** Configurable, math-driven transformation pipeline with an integrated Exponential Moving Average (EMA) filter for noise reduction.
- **LED Patterns:** (WIP) Dynamic visual feedback utilizing Programmable I/O (PIO).
- **Intra-communication:** (WIP) High-speed data transfer between split keyboard halves.
- **USB HID:** (WIP) Standardized host-computer interfacing.

Hardware Stack

- Microcontroller: Raspberry Pi Pico 2 / RP2350 (QFN-80 Package)
- C/C++ Pico SDK

Folder Structure

- [src/](#): Contains the main application logic and entry point.
- [src/](#): Contains actively developed, project-specific hardware and logical modules (e.g., joystick, leds, comm).
- [include/](#): Global headers and hardware pinout definitions.
- [docs/](#): Project tracking, change logs, and architectural planning.
- [tests/](#): Unit testing for independent modules.

Getting Started

1. Install the Pico C/C++ SDK.
2. Clone this repository.
3. Build + Upload and Monitor on PlatformIO